

```
pip install mysql-connector-python
```

```
CREATE DATABASE studentdb;
```

```
USE studentdb;
```

```
CREATE TABLE students (  
    id INT PRIMARY KEY,  
    name VARCHAR(50),  
    age INT,  
    department VARCHAR(50)  
);
```

```
import mysql.connector
```

```
# Connect to MySQL
```

```
db = mysql.connector.connect(  
    host="localhost",  
    user="root",  
    password="your_password",  
    database="studentdb"  
)
```

```
cursor = db.cursor()
```

```
# ----- CREATE -----  
def add_student():  
    id = int(input("Enter ID: "))
```

```
name = input("Enter Name: ")
age = int(input("Enter Age: "))
dept = input("Enter Department: ")
```

```
sql = "INSERT INTO students VALUES (%s, %s, %s, %s)"
values = (id, name, age, dept)
```

```
cursor.execute(sql, values)
db.commit()
```

```
print("Student Added Successfully!\n")
```

```
# ----- READ -----
```

```
def view_students():
    cursor.execute("SELECT * FROM students")
    records = cursor.fetchall()
```

```
    print("\nStudent Records")
    print("-----")
    for row in records:
        print(row)
    print()
```

```
# ----- UPDATE -----
```

```
def update_student():
    id = int(input("Enter Student ID: "))
    new_dept = input("Enter New Department: ")
```

```
    sql = "UPDATE students SET department=%s WHERE
id=%s"
    values = (new_dept, id)
```

```
    cursor.execute(sql, values)
    db.commit()
```

```
    print("Student Updated Successfully!\n")
```

```
# ----- DELETE -----  
def delete_student():  
    id = int(input("Enter Student ID: "))  
  
    sql = "DELETE FROM students WHERE id=%s"  
  
    cursor.execute(sql, (id,))  
    db.commit()  
  
    print("Student Deleted Successfully!\n")
```

```
# ----- MENU -----  
while True:  
    print("===== STUDENT MANAGEMENT =====")  
    print("1. Add Student")  
    print("2. View Students")  
    print("3. Update Student")  
    print("4. Delete Student")  
    print("5. Exit")  
  
    choice = input("Enter Choice: ")  
  
    if choice == "1":  
        add_student()  
  
    elif choice == "2":  
        view_students()  
  
    elif choice == "3":  
        update_student()  
  
    elif choice == "4":  
        delete_student()  
  
    elif choice == "5":
```

```
print("Program Ended.")  
break
```

```
else:  
    print("Invalid Choice!\n")
```

```
# Close connection  
cursor.close()  
db.close()
```